Problem 1.

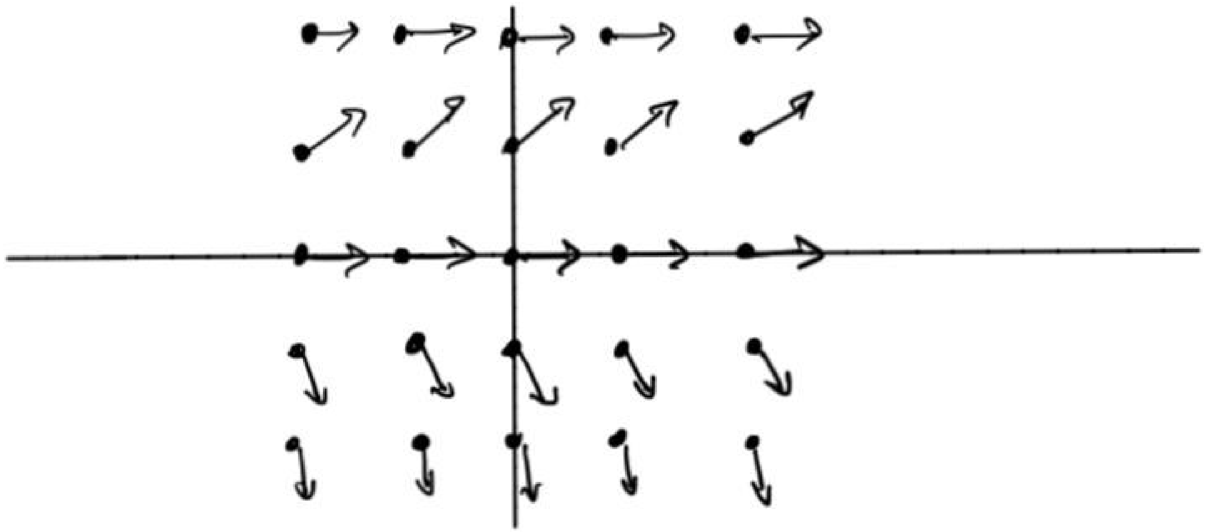
$$y' = 2y - y^2$$

point	slope
$(0,0)$	0
$(0,1)$	1
$(0,2)$	0
$(1,0)$	0
$(-1,2)$	0
$(-2,-2)$	-8

point	slope
$(1,1)$	1
$(1,2)$	0
$(2,0)$	0
$(2,1)$	1
$(-1,-1)$	-3
$(-2,1)$	1

point	slope
$(2,2)$	0
$(-1,0)$	0
$(-2,0)$	0
$(1,-1)$	-3
$(-1,-2)$	-8
$(-2,2)$	0

point	slope
$(1,-2)$	-8
$(2,-1)$	-3
$(2,-2)$	-8
$(-1,1)$	1
$(-2,-1)$	-3



Problem 2. $x_0 = 0, \quad y_0 = 1, \quad h = 0.1, \quad f(x,y) = y$

$x_1 = x_0 + h = 0.1, \quad y_1 = y_0 + f(x_0, y_0)h = 1 + 1 \times 0.1 = 1.1$

$x_2 = x_1 + h = 0.2, \quad y_2 = y_1 + f(x_1, y_1)h = 1.1 + 1.1 \times 0.1 = 1.21$

$x_3 = x_2 + h = 0.3, \quad y_3 = y_2 + f(x_2, y_2)h = 1.21 + 1.21 \times 0.1 = 1.331$

Q3.

Start with the differential eq$^{\underline{n}}$ for the concentration.

$$c'(t) = -K\,c(t)$$

Divide both sides by c(t):

$$\frac{c'(t)}{c(t)} = -K.$$

After solving this, we get

$$\ln c = -Kt + D$$

$$\Rightarrow c(t) = e^{D-Kt}, \quad D \text{ is any constant.}$$

At $t=0$, $c(0) = c_0 \Rightarrow c(0) = e^{D}$

$$\Rightarrow c(t) = c_0\,e^{-Kt}.$$

$\square$

Q4.

Given $(xy^2 - x^2)\,dx + (3x^2y^2 + x^2y - 2x^3 + y^2)\,dy = 0$ — (1)

Here

$M = xy^2 - x^2$ and $N = 3x^2y^2 + x^2y - 2x^3 + y^2$

$\therefore \dfrac{\partial M}{\partial y} = 2xy$ and $\dfrac{\partial N}{\partial x} = 6xy^2 + 2xy - 6x^2$

$$\therefore \frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) =$$

$$\frac{1}{xy^2 - x^2}\left(6xy^2 + \cancel{2xy} - 6x^2 - \cancel{2xy}\right)$$

$$= \frac{6x(y^2 - x)}{x(y^2 - x)} = 6$$

which being a constant can be treated as a function of y alone.

$\therefore$ IF of (1) : $= e^{\int 6\,dy} = e^{6y}$

Multiplying (1) by I.F :

$$e^{6y}(xy^2 - x^2)\,dx + e^{6y}(3x^2y^2 + x^2y - 2x^3 + y^2)\,dy = 0$$

whose sol$\underline{n}$ is:

$$\int e^{6y}\,(xy^2 - x^2)\,dx + \int e^{6y}\,y^2\,dy = c$$

(Treating 'y' as constant)

or $$e^{6y}\left[\left(\tfrac{1}{2}\right)x^2y^2 - \left(\tfrac{1}{3}\right)x^3\right] +$$

$$y^2\left(\tfrac{1}{6}\right)e^{6y} - \int (2y)\left(\tfrac{1}{6}\right)e^{6y}\,dy = 0$$

(Fill the gaps in intermediate steps)

or $$e^{6y}\left[\frac{1}{2}x^2y^2 - \frac{1}{3}x^3 + \frac{1}{6}y^2 - \frac{1}{18}y + \frac{1}{108}\right] = c,$$

c being an arbitrary constant.

$\square$

Q5. Rewriting;

$$\frac{dx}{dy} = x^2y^3 + xy$$

or $\frac{dx}{dy} - yx = y^3x^2$

or $x^{-2}\left(\frac{dx}{dy}\right) - x^{-1}y = y^3$ — (1)

Put, $x^{-1} = v$ and $-x^{-2}\left(\frac{dx}{dy}\right) = \frac{dv}{dy}$,

(1) becomes;

$$-\left(\frac{dv}{dy}\right) - yv = y^3,$$

or $\frac{dv}{dy} + yv = -y^3$, which is

linear in v and y and

$I.F = e^{\int y\,dy} = e^{y^2/2}$ and it's sol^n

is

$$v \cdot e^{y^2/2} = -\int y^3 e^{y^2/2}\,dy + C$$

$$= -2\int t e^t\,dt + C$$

(Putting $y^2/2 = t$ & $y\,dy = dt$)

$$= -2\left[t \cdot e^t - \int 1 \cdot e^t \, dt + c\right]$$

$$= -2\,(t e^t - e^t) + c$$

$$\Rightarrow \quad x^{-1} e^{y^2/2} = -2 e^{y^2/2} \left(\frac{y^2}{2} - 1\right) + c,$$

as $v = x^{-1}$.

$\therefore$ Required solⁿ is

$$\boxed{\frac{1}{x} = 2 - y^2 + c e^{-y^2/2}}$$, c being an arbitrary constant.